

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0495

Cwec Iola 345 kV
July 23, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0495. The Interconnection Customer proposes a 303.4 MW natural gas combined cycle facility interconnecting at the Cwec Iola 345 kV substation in the Pennsylvania area. The request was received on July 04, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 283 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,187,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Tuco / John Day 138 KV line	Firstenergy Fort Martin Power Sta / Cwec Iola 345 KV	123.0	231	23.4
Matanzas / John Day 138 KV line	Cwec Iola / Firstenergy Ft Martin Power Sta 345 KV	122.1	240	23.4
Auburn Road / Prides Corner 345 KV	Firstenergy Ft Martin Power Sta / Ronco 345 KV L	116.1	1,244	60.0
Auburn Rd / Westward 138 KV	Prides Corner / Auburn Road 345 KV	115.8	201	5.7
Lakehead Naubinway / Medora 138 KV L	Cwec Iola / Firstenergy Ft Martin Power Sta 345 KV	106.5	269	7.8
Westward Substation / Cavendish 138 KV	Auburn Rd / Prides Corner 345 KV	106.1	226	5.7
Auburn Rd 500/345 kV Transformer Bank	Acec Dellwood / Auburn Rd 345 KV L	104.8	676	24.1
Lakehead Naubinway / Spring Vly 138 KV line	Cwec Iola / Firstenergy Fort Martin Power Station 345 KV	101.8	261	7.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tuco / John Day 138 KV line	Rebuild line with higher-capacity conductor	\$2,036,000	19% (\$388,000)
Matanzas / John Day 138 KV	Rebuild line with higher-capacity conductor	\$2,766,000	19% (\$527,000)
Auburn Road / Prides Corner 345 KV	Rebuild line with higher-capacity conductor	\$11,137,000	19% (\$2,071,000)
Westward / Auburn Rd 138 KV Line	Rebuild line with higher-capacity conductor	\$8,017,000	10% (\$795,000)
Lakehead Naubinway / Medora 138 KV	Reconductor line	\$1,792,000	18% (\$323,000)
Westward Substation / Cavendish 138 KV	Reconductor line	\$1,818,000	10% (\$180,000)
Aubur Rd 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$36,414,000	16% (\$5,842,000)
Lakehead Naubinway / Spring Vly 138 KV line	Reconductor line	\$339,000	18% (\$61,000)
	Total allocated to project		\$10,187,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.